

Talon Dynamics LLC

White Paper in Response to Request for Information FA8629-26-RFI-4ISR

Prepared for the Air Force Life Cycle Management Center, ISR & SOF Directorate

18 May 2026 | POC: Marcus Webb, Director of BD — m.webb@talondyn.example — (321) 555-0177

Platform: TD-9 Vagrant expendable-leaning attritable UAS

Overview

Talon Dynamics is pleased to submit this white paper describing the TD-9 Vagrant, a deliberately minimal, low-cost attritable ISR aircraft engineered to sit at the expendable end of the attritable/reusable continuum described in the RFI. Our design philosophy trades objective-level features for radical affordability and producibility. The Vagrant is intended to be procured in quantity, flown hard, and written off without ceremony.

Aircraft Performance

The Vagrant delivers a mission radius of 115 nautical miles from the launch and recovery site to the collection area, with a time on station of 4.5 hours at best-endurance speed, not counting transit. Typical cruise is conducted at 8,500 feet above mean sea level, with a service ceiling of 14,000 feet. The aircraft climbs at approximately 900 feet per minute at gross weight and reaches a maximum airspeed of 95 knots true. These figures were validated across 212 flight hours of developmental testing at our Florida range. The airframe is rated for operations in light precipitation and sustained surface winds of 20 knots. We do not claim all-weather capability: the Vagrant is a fair-weather mass asset by design, and we believe this trade is consistent with the RFI's encouragement of solutions that deviate from objectives where doing so materially improves affordability and producibility.

The complete system - two aircraft, ground station, and launch equipment - packs into a single pickup truck bed or one 463L pallet position quarter. Two operators can be airborne within 30 minutes of arrival at an unimproved site. Launch is by elastic catapult; recovery is by deep-stall belly landing on any 50-meter clear area. Rapid deployment to austere locations is the cornerstone of the design.

Avionics, Communications, Navigation, and C2

The Vagrant provides mission-essential communications: an encrypted line-of-sight datalink carrying FMV and telemetry to approximately 60 nautical miles. Beyond-line-of-sight operation is achieved not by SATCOM but by autonomous execution of the pre-programmed mission plan, with onboard recording for post-mission exploitation; the aircraft flies its route, collects, and returns. We consider this fully responsive to the command and control requirement, which permits autopilot pre-programmed missions as an alternative to secure BLOS links. Link-16 is not offered on this platform.

Navigation is mission-essential: uncrewed-aircraft-grade GPS with a MEMS inertial unit for short-duration GPS outage ride-through. We do not offer an alternative PNT suite at this price point. The aircraft provides electrical power and standard interfaces for its sensor, autopilot, communications, and navigation subsystems as an integrated whole; payload power available is 150 watts at 12 volts DC.

Mission planning is performed in our WaypointOne desktop application, which supports route planning with terrain awareness. Automated planning, dynamic in-flight re-tasking, sensor scheduling, and air-traffic deconfliction features are not included. The Vagrant carries no onboard processing beyond flight-critical computing; all exploitation is performed post-flight at the ground station.

Ground Control Station

The ground control station is a single ruggedized tablet with an antenna mast, total weight 6 kilograms, providing mission-essential data processing (moving map, FMV display, and recording). The GCS can be positioned separately from the launch and recovery element. It integrates with the aircraft, the sensor

payload, our C2 datalink, the navigation solution, and exports imagery to Android Team Awareness Kit end user devices over a local mesh radio. The collaborative analysis environment and automated report dissemination described as objectives are not offered; we would note these functions are better hosted at the exploitation enterprise level than on an attritable system's field GCS.

Sensor Payloads

The baseline payload is a stabilized gimbal with full 360-degree azimuth coverage carrying an electro-optical daylight camera and an uncooled long-wave infrared imager. Video is standard definition; ground sample distance is approximately 20 centimeters at a slant range of 2 kilometers. All mission imagery is stored onboard on a 512 gigabyte removable card until transferred to the end user device, and the sensor is integrated with the aircraft, navigation (for frame geotagging), and EUD download paths both during and after the mission. Video is processed and recorded at 50 megabits per second in accordance with the stated requirement.

The Vagrant does not offer moving target indication, target tracking, a laser range finder, or a laser target indicator. A signal detection payload is available as an alternate nose: a software-defined receiver covering 100 MHz to 3 GHz that detects and geolocates push-to-talk emitters; waveform particulars can be discussed at the appropriate classification. This payload integrates with the aircraft, C3, navigation, and EUD in the same manner as the imaging payload. No sensors beyond those described are offered.

Key System Attributes

Talon Dynamics is pursuing a USAF Group 2 airworthiness assessment with an anticipated completion in Q4 2026; the design conforms to the applicable MIL-HDBK-516C criteria and we will meet the Group Type standard. An IFF Mode 3 transponder is fitted as standard equipment. Ground segment components are compatible with connection to secure networks through government-furnished cross-domain solutions. Production capacity today is 40 aircraft per month from our Melbourne, Florida facility, scalable to 120 per month within 90 days of contract start; first deliveries within 30 days after receipt of order.

Affordability Statement

Unit fly-away price at a 500-unit order quantity is under \$95,000. We contend that four Vagrants on station provide greater operational resilience than one exquisite platform at eight times the cost, and we welcome the opportunity to discuss this trade with the Capabilities Development Office.